Tanmay Bhuskute

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Cincinnati, OH

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, C/C++, SQL, Go.
Web & Backend: React, Node.js, Express.js, Django, Flask, FastAPI, REST APIs, GraphQL, Microservices.
Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Elasticsearch.
Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, GitLab CI/CD, Jenkins, Git, Linux, ArgoCD, Helm.
Testing & Practices: JUnit, Pytest, Jest, SonarQube, Integration Testing, Agile/Scrum, Design Patterns.
AI & Data: Gemini, Vertex AI, PyTorch, TensorFlow, scikit-learn, LangChain, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline that reproduces Android GUI bugs from screen recordings, evaluated against the **ViBR** baseline across 53 runs on 20+ real-world Android apps using Gemini 2.5 Pro and Vertex AI.
- Built Python harness (uiautomator2, ADB) with stall detection, achieving **25.7x latency reduction** over baseline.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** workflows with Python (Boto3), replacing legacy Ruby scripts and cutting deployment time **40%**.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%**.
- Executed **44 production environment upgrades** in 4 months, diagnosing live issues across Linux, EC2, and databases.
- Quality Coach and SonarQube Specialist: raised automated code coverage **35% to 78%**; mentored **5 software engineers**.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built full-stack **React and MySQL** application digitizing departmental records and deployed on-premises MySQL with authentication and permissions, cutting document retrieval time by **90%** (5 min to 30 sec).

PROJECTS AND RESEARCH

NYC Mobility Forecasting [Python, PySpark, FastAPI, PostgreSQL, Redis] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** dataset); two-stage Poisson hurdle model improved MAPE **18.77% to 16.52%**, served via FastAPI with PostgreSQL and Redis caching.

PolicyPilot: Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Built full-stack TypeScript app (React, Express 5 REST API, MCP server) with Zod schema validation at every data boundary.
- Designed **17-tool MCP server** exposing structured LLM tool-calling for PA extraction, patient-readiness evaluation, and cross-payer comparison, consumable by external AI agents.

Spendora: AI-Powered Personal Finance Platform [Next.js, TypeScript, PostgreSQL, Gemini] Jun 2025

- Engineered Gemini-powered, schema-validated transaction workflows for natural-language, voice, and receipt inputs, reducing median expense-entry time by **72%** (29s to 8s).
- Implemented Supabase auth, PostgreSQL row-level security, and AI usage quotas, covered by **80 test scenarios**.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).